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CARAPACE ASYMMETRIES IN CRABS

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ABSTRACT

Through the geometric morphometric approach of the landmark-based method applied on adult right-handed specimens of *Carcinus aestuarii*, *Liocarcinus depurator* and *Potamon fluviatile*, statistical analyses showed significant outcomes for fluctuating asymmetry in the three studied species, and directional asymmetry only in *P. fluviatile* (that showed a less convex carapace shape on the right side, where larger chela occurred). Although these findings may be correlated to some environmental constraints, such as soil composition, refuge occurrence and predatory pressure, also other interpretations may be proposed.

Key words. — *Carcinus aestuarii*, *Liocarcinus depurator*, *Potamon fluviatile*, fluctuating asymmetry, directional asymmetry

RIASSUNTO

Attraverso la morfometria geometrica ed il metodo basato sui landmarks, applicati ad esemplari adulti e destrorsi di *Carcinus aestuarii*, *Liocarcinus depurator* e *Potamon fluviatile*, le analisi statistiche hanno significativamente evidenziato un'asimmetria fluttuante nelle tre specie studiate e un'asimmetria direzionale solo in *P. fluviatile* (che mostra una forma del carapace meno convessa sul lato destro, dove si trova la chela più grande). Sebbene questi risultati possano correlare con alcuni parametri ambientali, come la composizione del suolo, la presenza di rifugi e la pressione predatoria, è possibile proporre alcune interpretazioni dei fenomeni osservati.

INTRODUCTION

The preferential use and more precise performance for body parts on one side, rather than the other, is called handedness and behavioural asymmetries may bring about morphological asymmetries (Palmer, 2004). Palmer hypothesized two main evolutionary routes to reach the different handedness in crabs, both

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starting from the normal condition for bilaterians that is "fluctuating asymmetry" (= FA = minor random deviations from bilateral symmetry in ancestors). FA may be visualized as the base on which natural selection acts, originating either in directional asymmetry (DA, that is asymmetric organisms of which the two extremes are positively selected), or antisymmetry (AS, the lack of symmetry when only one side is positively selected). While FA is used to measure developmental instability (as indicator of stress in organisms inhabiting disturbed environments: Graham et al., 1993; Van Dongen, 2006; Lezcano et al., 2015; Scalici et al., 2015, 2017), DA and AS are developmentally controlled (Van Valen, 1962; Palmer & Strobeck, 1992).

Heterochely (i.e., asymmetry of claws) in crabs, understood as the morphological result of their handedness, represents a poorly understood phenomenon (Abby-Kalio & Warner, 1989; Daniels, 2001; Scalici & Gherardi, 2008) in spite of their role on the adaptive strategies and radiation of brachyurans (Hartnoll, 1978, 1982; Vannini et al., 1983; Daniels, 2001; Duarte et al., 2008; Juanes et al., 2008; Scalici et al., 2013; Spani & Scalici, 2016). In the view that different body regions develop, function, and evolve jointly as a part of an entire functional system (Klingenberg, 2008, 2013), heterochely in crabs may affect other body parts as a consequence of their acquired functional handedness.

The main purpose of this study was to assess if heterochely may represent a morphological constraint driving modifications in other parts, particularly the carapace, and if the same phenomenon (heterochely) could affect crab species inhabiting different habitats in different ways. We approached this topic by exploiting the geometric morphometric techniques (Savriama & Klingenberg, 2011), since the decapod exoskeleton represents an adequate biological model to apply established landmark-based methods (Rufino et al., 2004). We chose to exploit three different heterochelous brachyurans, inhabiting lagoons, the sea, and rivers, respectively.

MATERIAL AND METHODS

The biological model organisms employed were: two species of Portunidae, viz., (1) the Mediterranean littoral crab *Carcinus aestuarii* Nardo, 1847, that inhabits brackish areas and coastal sea habitats characterized by rocky shores, and also salt marshes with abundant algal vegetation (Mistri et al., 2001); and (2) the sandy swimming crab *Liocarcinus depurator* (Linnaeus, 1758), occurring in intertidal areas with a sandy substrate (Freire, 1996); as well as one species of Potamidae, i.e., (3) the river crab *Potamon fluviatile* (Herbst, 1785), a tunneling species (Vannini & Gherardi, 1981), adapted to banks of lakes and rivers of variable

granulometry, mixed to sand-with-stones (authors' unpublished data). Significant heterochely has already been established in *L. depurator* and *P. fluviatile* by applying traditional morphometrics (Abelló et al., 1990; Micheli et al., 1990; Freire et al., 1996). Concerning *C. aestuarii*, a few studies have highlighted the heterochely in some other species of the same genus (Abby-Kalio & Warner, 1989; Silva & Paula, 2008) but none focusing on this species of portunid. So, the present authors were able to detect a significant heterochely in both size and shape of the claws in this species, by applying geometric morphometrics (Spani et al., in prep.).

In order to properly document the external morphology of the crabs, we photographed them, using a Canon Eos 700 D digital photographic camera, mounted on a support at 20 cm away from the carapace to limit spherical distortion and parallax problems. The photographs taken, always in duplicate, thus documented the carapace in dorsal view of 60 right-handed adults (30 females + 30 males) for each species investigated.

C. aestuarii and *L. depurator* were caught by professional fishermen from Venice Lagoon, northeastern Italy (33T, 5035674.02m N 2939778.08m E) and from Passoscuro, central Italy (33T, 4642337.75m N 264237.04m E), respectively. *P. fluviatile* were taken from the laboratory collection, since the species has drastically decreased in abundance within the national territory during the last decade (Barbaresi et al., 2007; Scalici et al., 2008). These specimens had previously been collected in the Rio Fiume, central Italy (32T, 4663078.26m N 745134.50m E).

Only right-handed crabs were studied, since left-handed individuals occurred in a very low frequency in each of the investigated species (less than 3%, pers. obs.). Moreover, individuals had to show no signs of cheliped regeneration whatsoever. To preserve all geometric features of the carapace, we used the landmark Cartesian coordinates (Zelditch et al., 2004), fixed twice as in fig. 1, using TpsDig (Rohlf, 2010). Double photographs and digitalization were performed for evaluating both sampling and measurement error, by exploiting the Procrustes analysis of variance (ANOVA) (Klingenberg, 2011). These errors all proved to be negligible for each sex of the three investigated species (P was always <0.01). Thus, the selected landmarks and configurations could reliably be used in all statistical analyses.

After fixation of the landmarks, the information on shape was extracted by removing all information on size, position, and orientation, and thus to standardize each specimen according to the configuration of a centroid with a full Procrustes fit (Dryden & Mardia, 1998), and representing the object symmetry of the studied structure (Klingenberg et al., 2002). We executed the Generalized Procrustes Superimposition (GPS) by scaling the configurations to unit centroid size (CS; see Bookstein, 1986), and overlapping the centroid of all configurations. Here, CS is a proxy of the size, computed as the square root of the sum of the squares of the distances of each landmark from the centroid (i.e., the mean point having as

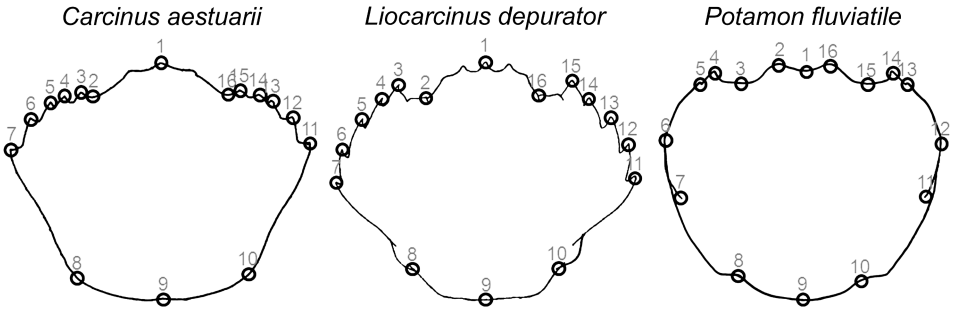


Fig. 1. Landmarks digitalized on the dorsal portion of the carapace in the three studied species of brachyuran crabs: *Carcinus aestuarii* Nardo, 1847, *Liocarcinus depurator* (Linnaeus, 1758) and *Potamon fluviatile* (Herbst, 1785).

abscissa and ordinate the mean of all the x and y coordinates of all the landmarks, respectively). Finally, we rotated each of the configurations around their centroid to allow finding the best fit of the corresponding landmarks.

DATA PROCESSING, RESULTS AND DISCUSSION

Before executing analyses, we evaluated if both eigenvalues and eigenvectors were stable despite low sample size, according to the procedures of Kocovsky et al. (2009). Specifically, we executed 1000 PCAs on 100 individuals randomly chosen from the investigated 180, selected the first three PC eigenvectors, and performed a regression analysis between the corresponding first 3 eigenvectors of PCA coming from the complete dataset. Then we calculated and considered the absolute value of the beta coefficient as a good measure of how these PCAs performed similarly in orienting the shape change. From the beta calculation emerged that PCAs were acceptably stable: mean(abs(beta1)) = 0.99235984; mean(abs(beta2)) = 0.8745298; mean(abs(beta3)) = 0.8132458. This means we may use the collected landmarks' matrices to further develop this study.

We first tested the sexual dimorphism for the carapace in both size and shape. Intraspecific sexual dimorphism was negligible for both CS (after Student's t-test: $t = 0.89$, $P = 0.21$ for *C. aestuarii*, $t = 1.63$, $P = 0.18$ for *L. depurator*, and $t = 1.01$, $P = 0.29$ for *P. fluviatile*) and shape variables (after the Procrustes analysis of variance ANOVA, with 1000 permutations: $F_{1,29} = 2.01$, $P = 0.17$ for *C. aestuarii*, $F_{1,29} = 1.45$, $P = 0.32$ for *L. depurator*, and $F_{1,29} = 2.79$, $P = 0.15$ for *P. fluviatile*) in the three species.

This allowed us to merge females and males in a combined data set for each of the investigated species. Specifically, once GPS was run, the aligned coordinates obtained by the configuration superimposition were entered into the

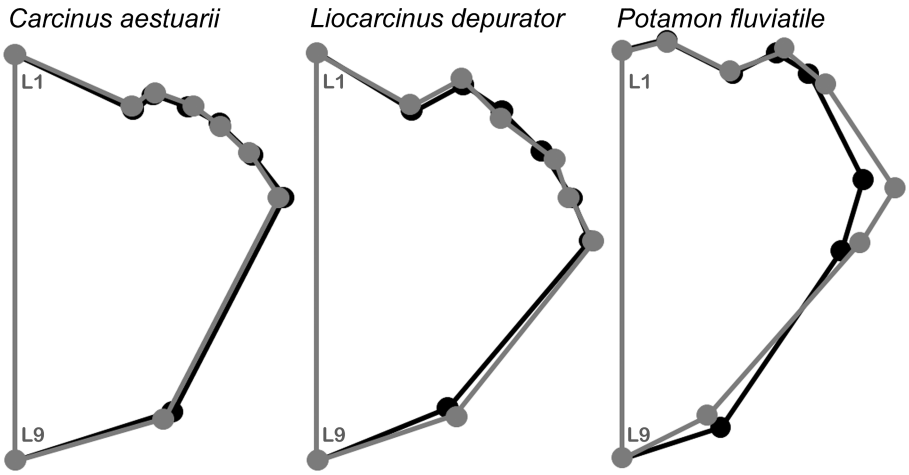


Fig. 2. Consensus of the left half carapace (gray) secularized on the right half (black) for the three investigated species, *Carcinus aestuarii* Nardo, 1847, *Liocarcinus depurator* (Linnaeus, 1758) and *Potamon fluviatile* (Herbst, 1785). Full lines represent the landmark links (L1 and L9 are landmark number 1 and landmark 9, respectively — see also fig. 1).

Procrustes ANOVA, with 1000 permutations (Klingenberg & McIntyre, 1998). This procedure was executed to analyse FA (measured by the side × individual interaction), DA (expressed by the main effect for sides), and the measurement error (ME, expressed by the residual term) in both CS and shape variables (that is the superimposed landmark coordinates) for each investigated species (see Klingenberg, 2011). The individuals and the carapace side were entered as random and fixed effects, respectively. The degrees of freedom are those for ordinary ANOVA, multiplied by the shape dimensions, which is twice the number of the landmarks minus the four degrees of freedom that are lost during superimposition (two degrees lost during translation, one degree for each of the two dimensions, and one each for scaling and rotation). To determine AS (Van Dongen et al., 1999), we visually examined scatter plots for each landmark of the left and right vectors after superimposition. The outcomes of the Procrustes ANOVA on both CS and shape variables indicated a significant effect of FA for the three studied species, and DA as the most important source of left-right shape differences only for PF (fig. 2). Specifically, FA was detected for centroid size in all species ($F = 7.201$, 9.774, 12.583 for *C. aestuarii*, *L. depurator*, *P. fluviatile*, respectively, P always <0.05) and for shape variables in *C. aestuarii* and *L. depurator* ($F = 8.814$, 11.115 for *C. aestuarii* and *L. depurator*, respectively, P always <0.05), while DA was observed only for shape variables in *P. fluviatile* ($F = 14.441$, $P < 0.05$). Since normal (rather than bimodal) distributions of differences were obtained, AS was discarded as a source of asymmetries.

With the advent of geometric morphometrics, significant directional asymmetry was highlighted in virtually every early study of shape asymmetry, and recognized as a subtle, but measurable and widespread phenomenon in the animal kingdom. Although, some output results on directional asymmetry were supported by relatively small subsamples, which raised questions about the real statistical significance of this method (see Klingenberg, 2015, and references therein). In this study, we performed statistical analyses by exploiting 60 individuals per species, avoiding any kind of issue related to small sample size and definitely demonstrating the robustness of our statistics. Nevertheless, given the strength of our statistics, this does not mean that the FA and DA observed in our specimens (representing only a population of each of the three investigated species) can be considered as a phenomenon that can be extended to the whole species.

We describe the carapace asymmetry as a passive consequence of mechanical features (regarding the acquired handedness, functional for several alimentary, reproductive, and fighting uses), being aware that we cannot explain the real evolutionary constraints driving both fluctuating and, overall, directional asymmetries, but that we may only hypothesize some scenarios regarding some of the possible driving forces that could be effective in the crabs' habitats.

Specifically, the main output of this investigation is the identification of geometric morphometric characters showing a carapace variability linked to both fluctuating and directional asymmetry, depending on the studied species. The present study clearly demonstrated the occurrence of fluctuating asymmetry in the carapace of the two studied portunids, while a directional asymmetry was established for the carapace of the river crab *P. fluviatile*. This adds a novel example of asymmetry in crabs, because, if on the one hand the asymmetry observed in the chelae is widespread among brachyurans (but not really quantified as a phenomenon, except for a few crab species, in particular those belonging to the genus *Uca*), on the other hand no information existed in literature for the asymmetric carapace shape, except for asymmetric anterolateral spines (Poore, 2004). Then it seems to be noteworthy that decapods may display directional asymmetry in other traits than claws. Although some ecological (Băncilă et al., 2010), physiological (Onken & McNamara, 2002), and behavioural (Tobo et al., 2012) reasons may be linked to the morphological asymmetry of the body in decapods, other selective pressures may play a possible role as well, such as habitat structure and heterogeneity (Idaszkin et al., 2013).

In fact, crabs in running waters (showing DA) inhabit rivers with heterogeneous bed structure, i.e., with a mix of sand, boulders and stones, and create burrows with narrow corridors therein. This is clearly a different way of life than that led by *C. aestuarii* and *L. depurator*, that exploit refuges among stones or between algae and other plants (pers. obs.). The asymmetry of the claws in freshwater

crabs is widely known and has been studied, and it was demonstrated that this has apparently evolved for different uses, such as predation, food manipulation, agonistic behaviour, and mate acquisition. In our study, however, we would like to focus not on the heterochely itself, but on the effect that an advantageous character, such as the asymmetry of the crab's claws, may have had on shaping the carapace of *P. fluviatile* in the course of evolution.

More studies ought to be carried out to confirm the observed pattern. Additionally, a much larger study involving more populations per species is required, considering the overall morphological variability of the crab. Indeed, in this respect, an extensive intraspecific phenotypic variability has been detected in brachyurans also within relatively restricted geographical areas (Brian, 2005; Idaszkin et al., 2013).

We are also aware that the functional and adaptive meanings of directional asymmetry are unclear in many taxa and it is plausible, for instance, that directional asymmetry in many external structures represents a non-adaptive consequence of developmental constraints relating to directional asymmetry of internal organs (Klingenberg, 2015). In the studied biological model organisms, since no evident internal asymmetries have ever been detected (pers. obs.), the directional asymmetry in the chelae driving asymmetry in the carapace shape in *P. fluviatile*, i.e., an adjacent external structure, seems to be a quite convincing idea.

Nonetheless, additional investigations about both heterochely and carapace shape may be addressed in more populations of various species, in order to understand as yet unknown ecological mechanisms driving the origin of asymmetry, in relation to functionality and evolution, and to evaluate the possibly different asymmetric trajectories in marine, brackish, and freshwater crabs.

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